

**Turiba University
(Jyoti)**

[Alogrithm , Flowchart]

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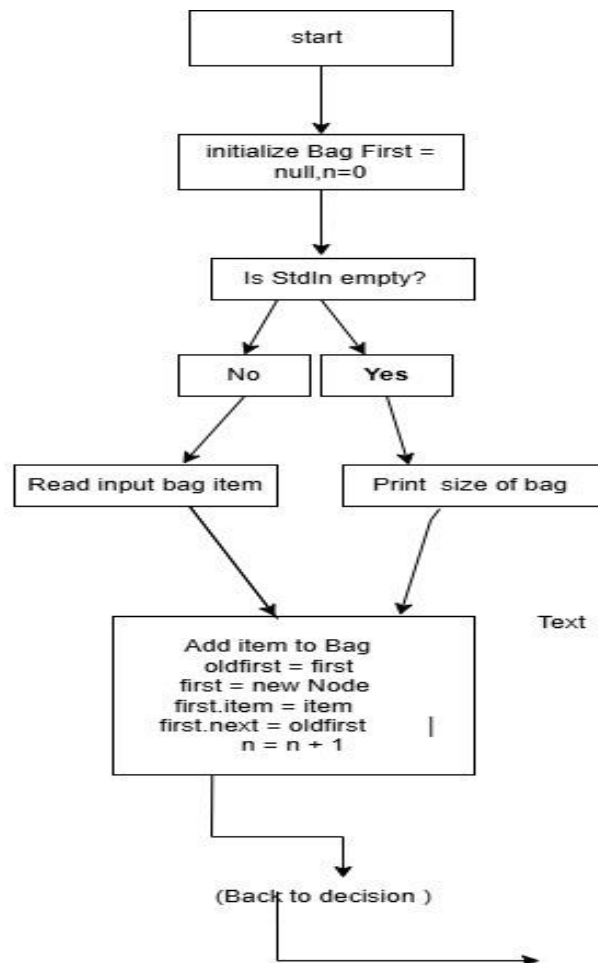
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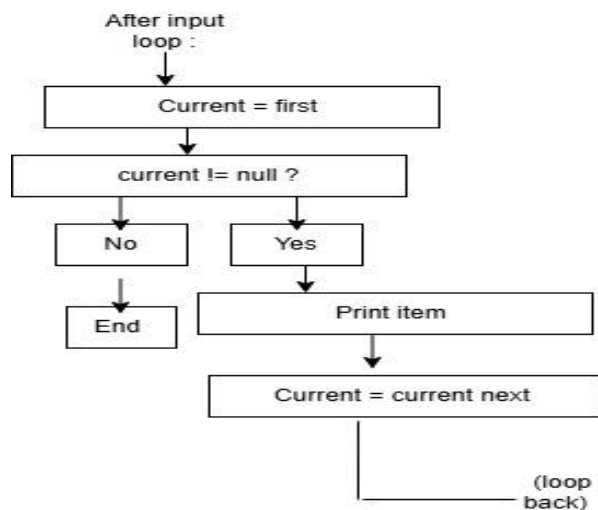
TOPIC

BAG.JAVA

FLOWCAHRT =



Text

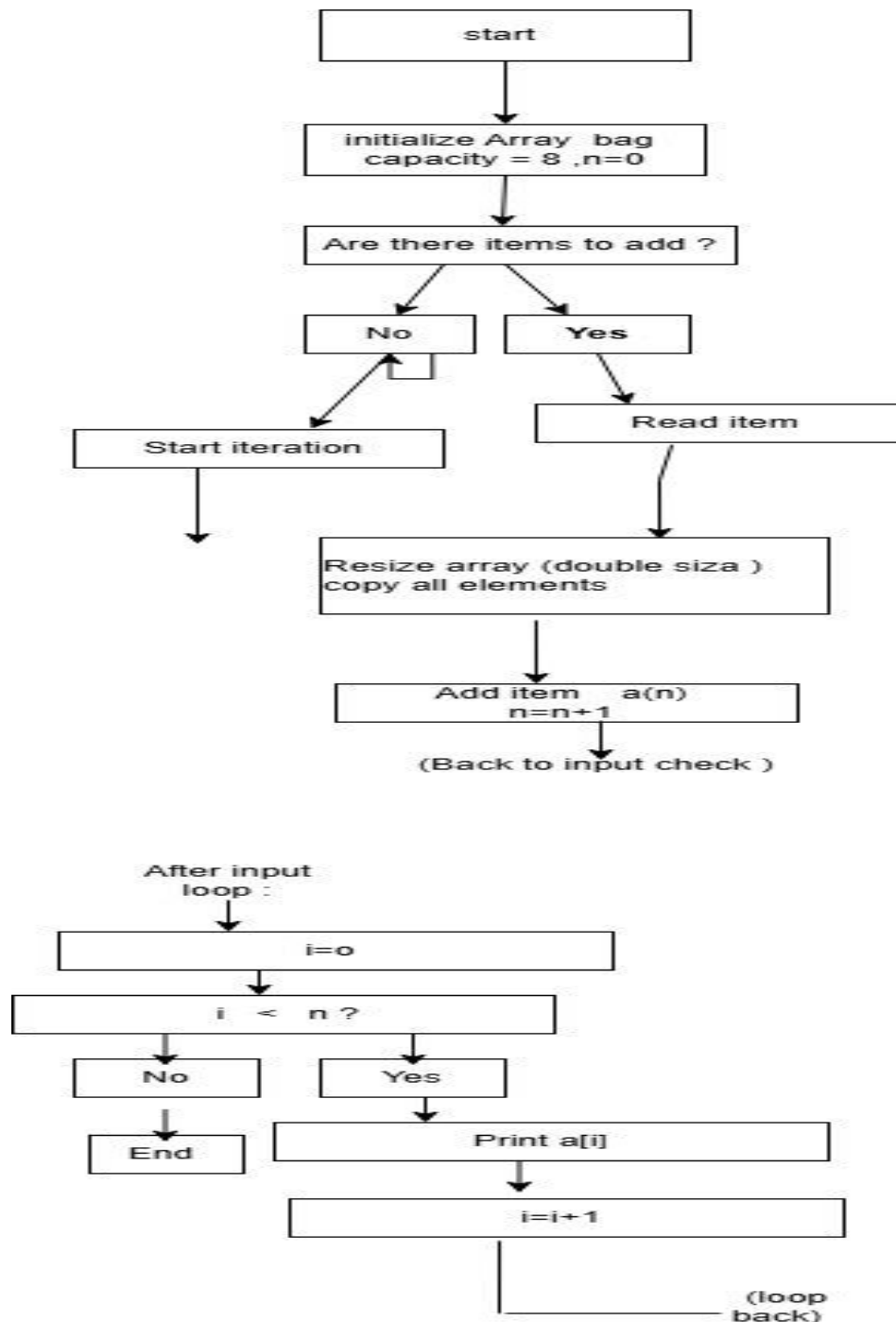


Algorithm

```
1. START
2. Initialize Bag
   first ← null
   n ← 0
3. WHILE input is not empty DO
   3.1 Read item
   3.2 Create new node
   3.3 Set new node.item ← item
   3.4 Set new node.next ← first
   3.5 Update first ← new node
   3.6 Increment n ← n + 1
   END WHILE
4. Print "size of bag = ", n
5. Set current ← first
6. WHILE current ≠ null DO
   6.1 Print current.item
   6.2 Set current ← current.next
   END WHILE
7. END
```

Second Topic

ResizingArrayBag.java



Algorithm

1. START
2. Initialize Bag

```
Create array a with capacity = 8
n ← 0
3. WHILE input/items are available DO
  3.1 Read item
  3.2 IF n == length of array THEN
    Create new array of size 2 × current size
    Copy all elements from old array to new array
    Replace old array with new array
  END IF
  3.3 Add item at index n → a[n]
  3.4 Increment n ← n + 1
END WHILE
4. Print all elements:
  Set i ← 0
  WHILE i < n DO
    Print a[i]
    i ← i + 1
  END WHILE
5. END
```

Thank you